API Reliability, Latency, Rate Limits, and Error Handling

FICTIONAL TRAINING DOCUMENT — designed for RAG/agent testing; no real company, policy, SLA, customer, or price is represented.

API baseline and objectives

Public APIs use HTTPS/JSON. Responses include X-Northstar-Request-Id for trace correlation. Capture request ID, endpoint, method, region and UTC timestamp before escalation.

Class	Typical p95	Investigate when
Read	<500 ms	>800 ms for 15 min
Write	<800 ms	>1.2 s for 15 min
Search	<1.5 s	>2.5 s for 15 min
Async submit	<700 ms	>1.0 s for 15 min

These are operational targets, not contractual SLAs unless explicitly contracted.

HTTP errors

Code	Interpretation / action
400	Validate schema, required fields, content type.
401	Token validity, audience, expiry, clock.
403	Workspace/project role and scope.
404	ID, region, visibility/permissions.
409	Refresh state; avoid duplicate operation.
429	Honor Retry-After; bounded exponential backoff + jitter.
500	Retry idempotent operation; capture request ID.
502/503	Gateway/dependency unavailable; backoff; check region health.
504	Upstream timeout; reduce scope if possible; retry.

Rate limits

Default fictional Growth limit: 600 requests/minute/workspace with burst allowance 60/10 seconds. Enterprise varies by contract/configuration. 429 may include Retry-After. Never blindly retry non-idempotent operations; use idempotency keys where supported.

Latency triage

• Compare p50/p95/error rate, not a single request.

• Check recent deployment/feature flag; DB pool saturation/slow queries; queue depth; cache hit rate; upstream dependency latency; regional network health.

• Determine blast radius: tenant, endpoint, region, or global.

Diagnostic headers

Header	Purpose
X-Northstar-Request-Id	Gateway/app/dependency trace correlation
X-Northstar-Region	Serving region
Retry-After	Retry delay after throttling/protection
X-RateLimit-Remaining	Approximate quota where exposed

Failure patterns

Latency up + DB connections saturated: investigate slow queries/pool/transactions. **503 immediately after one-region deploy:** compare previous release; rollback may be appropriate. **Only one tenant gets 429:** inspect tenant traffic/concurrency/contracted limit before declaring outage.

Escalate when

• Persistent multi-region latency above trigger >15 min.

• Critical API error rate >5% for 5 min.

• Broad authentication/checkout unavailability.

• Suspected data corruption or duplicate side effects.

• Impact meets Sev 1/2.

Retry safety

Retry 429/502/503/504 with bounded exponential backoff and jitter. For create/payment-like operations, use documented idempotency behavior. Avoid retry storms; cap attempts and concurrency.

Customer-safe response

Share public limits, request IDs, verified status and remediation. Do not expose internal cluster names, credentials, private IPs, raw security data, or unreviewed root-cause speculation.